

Lab 08

Due March 24 5:00 PM PST

STAT240 Spring 2026

2025-03-20

For this homework, provide a single rendered Quarto Markdown file in pdf format on crowdmark for the problems (you may render the QMD file to html, and then convert the html file to pdf using the print function on your web browser). You can use the template `Lab_07.qmd` from last week. **Provide in the markdown the final version of all of the code you wrote for this homework, and make sure long lines of code are wrapped in the rendered pdf. Lines which are cut off cannot be graded. If your Markdown file involves examining a large dataset, do not print the entire dataset to the markdown file in the steps of your solution (instead, suppress the output, or only show a small section of the data as an example).**

Problem 1

In these problems we will use the text of Jane Austen novels, which is included in the R package `janeaustenr`. We will also manipulate this text using the following code, which requires the `tidyverse` and the `tidytext` packages.

```
library(tidytext)
library(janeaustenr)

book_words <- austen_books() |>
  unnest_tokens(word, text)
```

Problem 1a

Looking only at the book “Sense and Sensibility”, determine the proportion of words in this text which contain 10 or more letters. You can include all words (for this text) that are contained

in `book_words`. Note that we did not cover a function to do this in class, so you should consult the documents for the `stringr` package available [here](#).

(3 points)

Problem 1b

Now compute this quantity for each of the books in `book_words`. Similarly, determine the year each book was published. Construct a plot where the x axis is the publication year and the y axis is the proportion of words with 10 or more letters for each of the books in `book_words`.

(5 points)

Hint: There are many ways you could get the year a book was published. For some of these books it may not be contained in this dataset.

Problem 1c

Now we will slightly modify the previous problem. For all books in this dataset, find all seven letter words which start with the last 4 letters in the alphabet and end with e. How many such unique words are there, and what are they?

(2 points)